USB Camera Intrinsic Parameters Calibration

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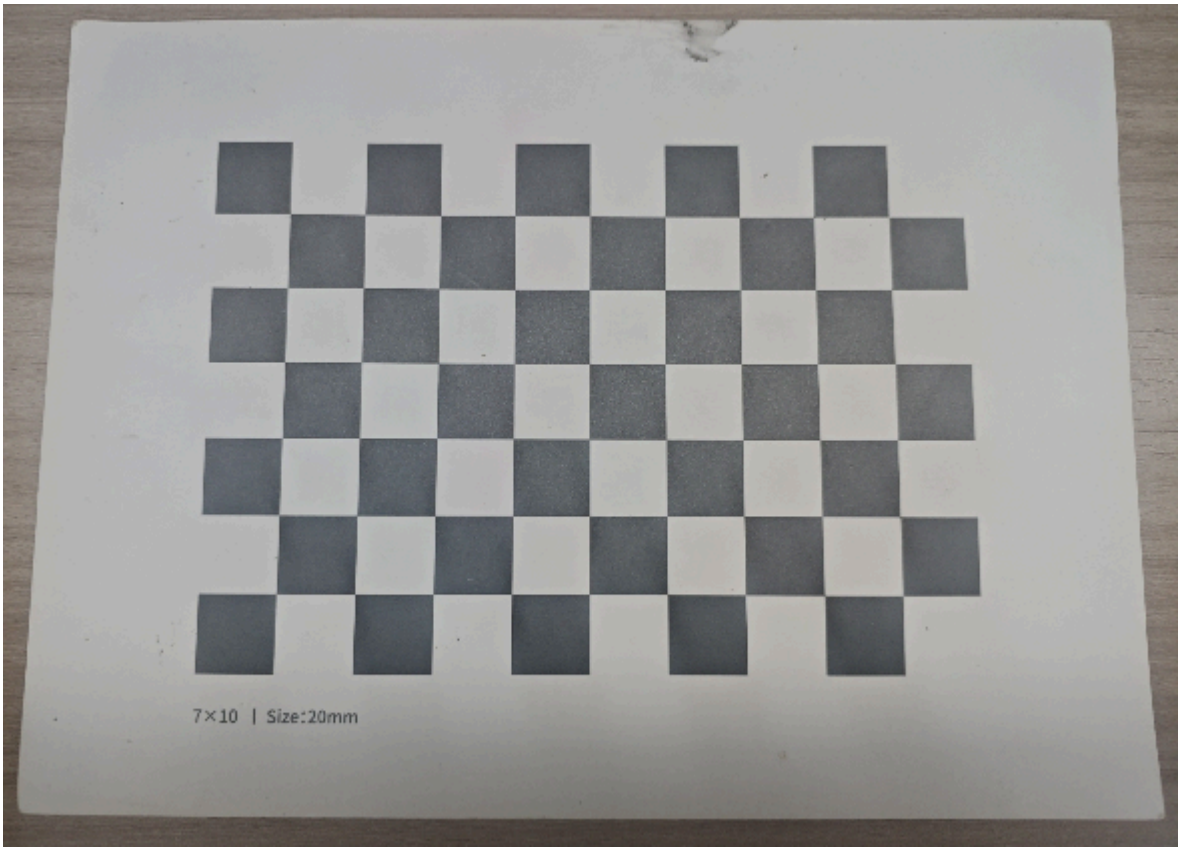
1. Course Content

- Calibrate the camera intrinsic parameters of the USB camera using a checkerboard calibration board, to be used for publishing the camera `camera_info` information

> **Important**
>
> - **This section is intended for users of the USB camera version**; depth camera users please skip this section
> - **Camera intrinsic parameters have already been calibrated in the factory image. Do not directly modify the factory camera intrinsic parameters file!** This section only teaches the camera intrinsic parameters calibration method and does not modify the factory camera intrinsic parameters. If you have development needs, you can modify and adjust the camera intrinsic parameters yourself.

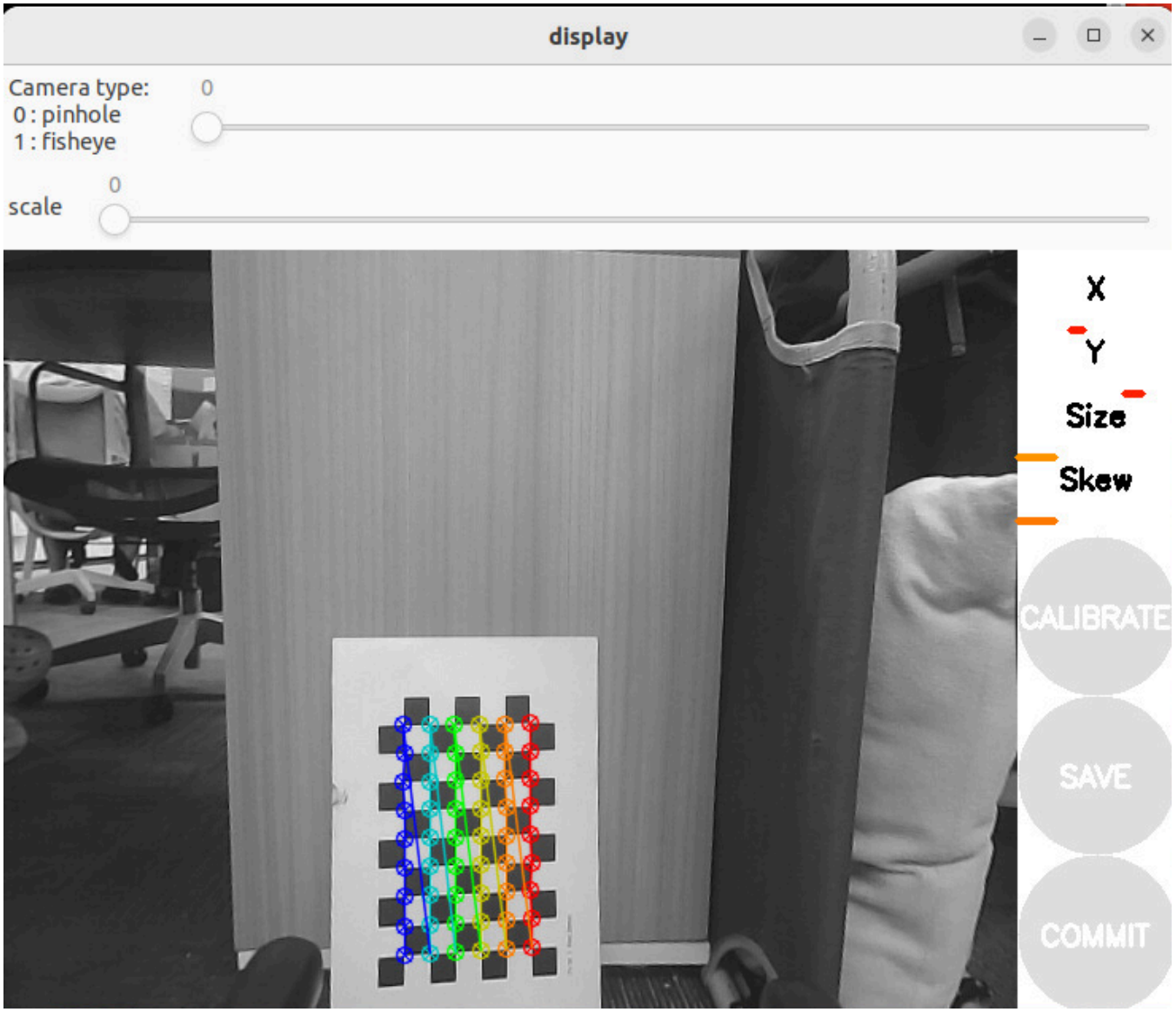
2. Preparation

- Prepare a checkerboard calibration board. From the accompanying materials [Attachments — Camera Checkerboard Calibration Board], download and print the calibration board for the corresponding camera and attach it to a hard cardboard or other flat surface



- Start the camera intrinsic parameters calibration program

```
ros2 launch yahboomcar_bringup camera_demo.launch.py demo:=camera_calibration
```



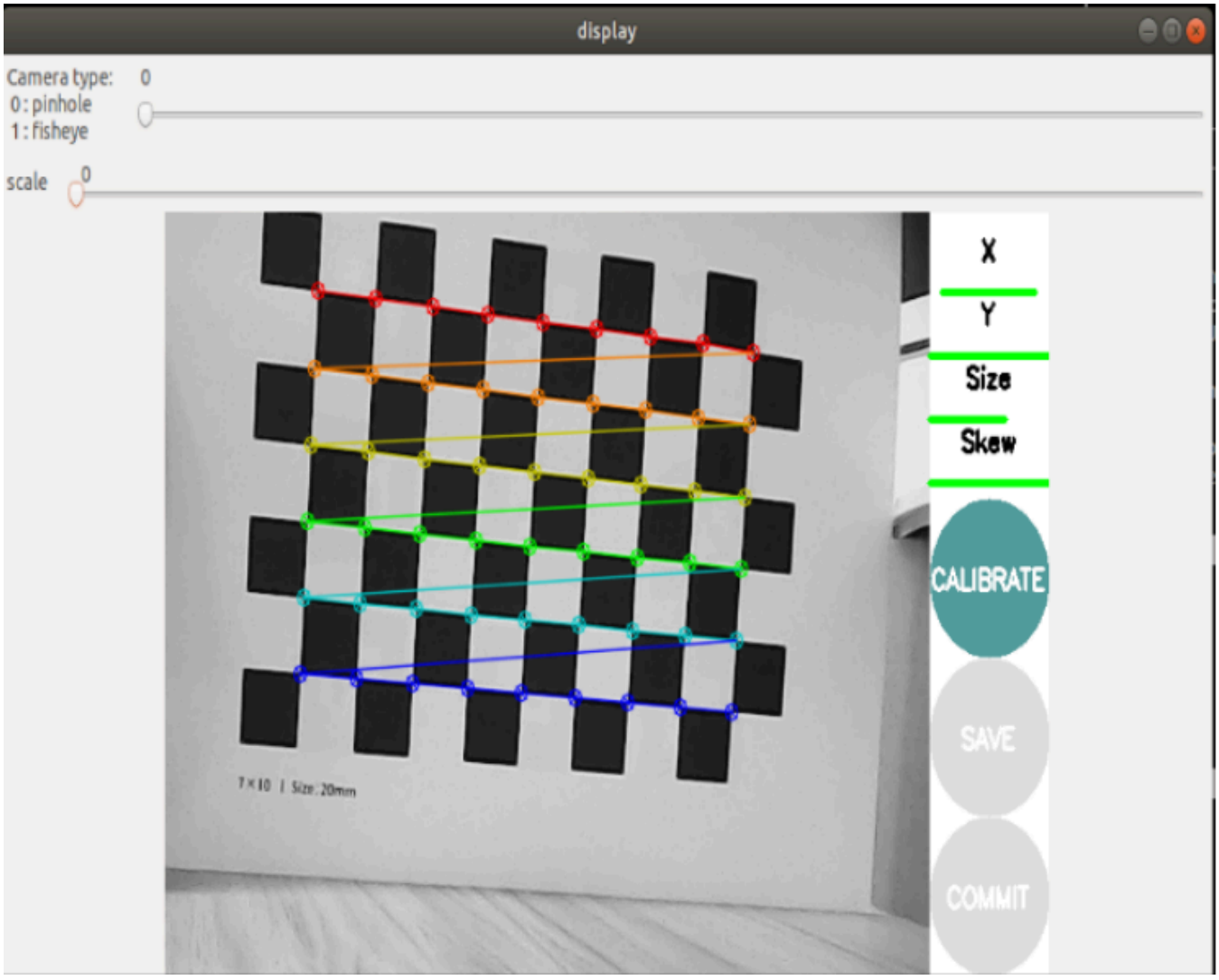
3. Calibration Case Demonstration

size: the number of inner corners of the calibration checkerboard, for example 9X6, meaning six rows and nine columns of corners in total.

square: the side length of the checkerboard, in meters.

Calibration Interface

- X: left-right movement of the checkerboard in the camera's field of view; Y: up-down movement of the checkerboard in the camera's field of view; Size: front-back movement of the checkerboard in the camera's field of view; Skew: tilted rotation of the checkerboard in the camera's field of view
- Place the checkerboard in the center of the image and change its pose. The system will recognize it automatically. The best case is that the lines below 【X】, 【Y】, 【Size】, and 【Skew】 first turn from red to yellow and then to green as data is collected, filling up as much as possible.

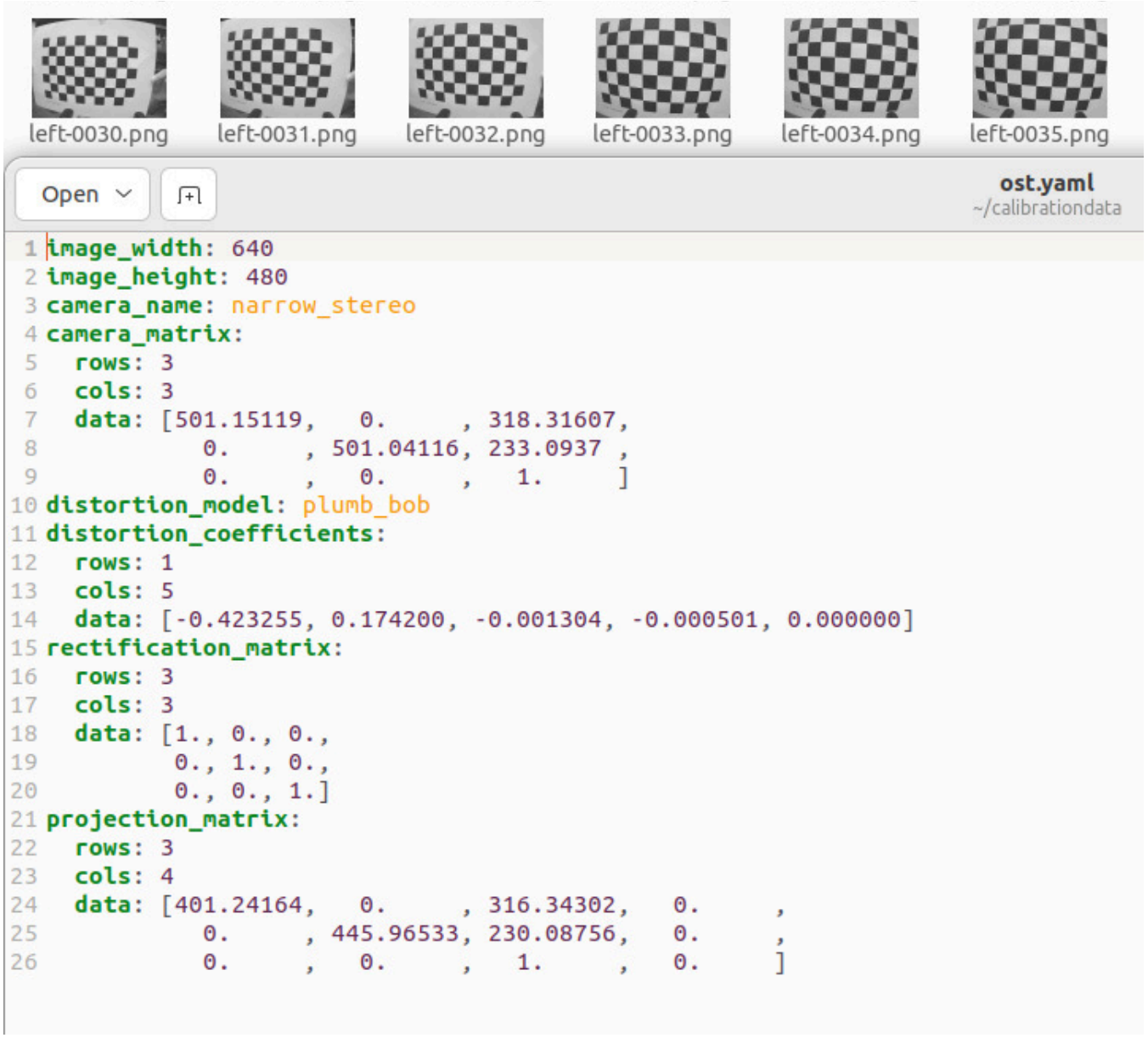


- Click 【CALIBRATE】 to calculate the camera intrinsic parameters. The more images, the longer it takes; just wait. (Sixty or seventy images is about enough; too many can easily cause it to freeze).
- Click 【SAVE】 to save the calibration result to 【/tmp/calibrationdata.tar.gz】 in the currently running terminal.
- After calibration is finished, you can move the 【/tmp/calibrationdata.tar.gz】 file to your user directory to view its contents

- Move and decompress the calibration file

```
sudo mv /tmp/calibrationdata.tar.gz ~
```

After decompression, it contains the images just calibrated, an ost.txt file, and an ost.yaml file. ost.yaml is the intrinsic parameters of the calibrated camera



4. Camera Intrinsic Parameters Explanation

- The path of the camera intrinsic parameters file actually loaded when the USB camera starts:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/param/usb_camera_info.yaml
```

- `image_width` / `image_height`: camera image resolution, here 640×480 pixels
- `camera_name`: camera name identifier
- `camera_matrix`: 3×3 camera intrinsic matrix K = [fx, 0, cx; 0, fy, cy; 0, 0, 1]
  - `fx`, `fy`: focal lengths in the x/y directions (in pixels), which determine the image scaling ratio
  - `cx`, `cy`: principal point coordinates (the pixel position of the optical center in the image), ideally close to the image center
- `distortion_model`: distortion model; `plumb_bob` is the 5-parameter distortion model (k1, k2, p1, p2,

k3)

- `distortion_coefficients`: distortion coefficients
  - `k1`, `k2`, `k3`: radial distortion coefficients, describing the barrel/pincushion distortion caused by lens curvature
  - `p1`, `p2`: tangential distortion coefficients, describing the distortion caused by lens mounting deviation
- `rectification_matrix`: 3×3 rectification matrix; for a monocular camera it is the identity matrix (no stereo rectification needed)
- `projection_matrix`: 3×4 projection matrix P = [fx', 0, cx', Tx; 0, fy', cy', Ty; 0, 0, 1, 0]
  - For a monocular camera Tx=0, Ty=0, used to project 3D points onto the 2D image plane

```
image_width: 640
image_height: 480
camera_name: default_cam
camera_matrix:
  rows: 3
  cols: 3
  data: [501.151190, 0.000000, 318.316070, 0.000000, 501.041160, 233.093700, 0.000000,
0.000000, 1.000000]
distortion_model: plumb_bob
distortion_coefficients:
  rows: 1
  cols: 5
  data: [-0.423255, 0.174200, -0.001304, -0.000501, 0.000000]
rectification_matrix:
  rows: 3
  cols: 3
  data: [1.000000, 0.000000, 0.000000, 0.000000, 1.000000, 0.000000, 0.000000, 0.000000,
1.000000]
projection_matrix:
  rows: 3
  cols: 4
  data: [401.241640, 0.000000, 316.343020, 0.000000, 0.000000, 445.965330, 230.087560,
0.000000, 0.000000, 0.000000, 1.000000, 0.000000]
```